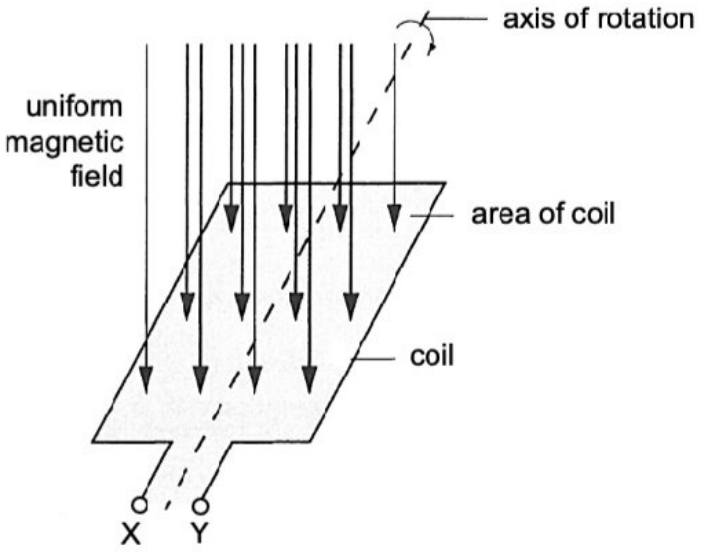
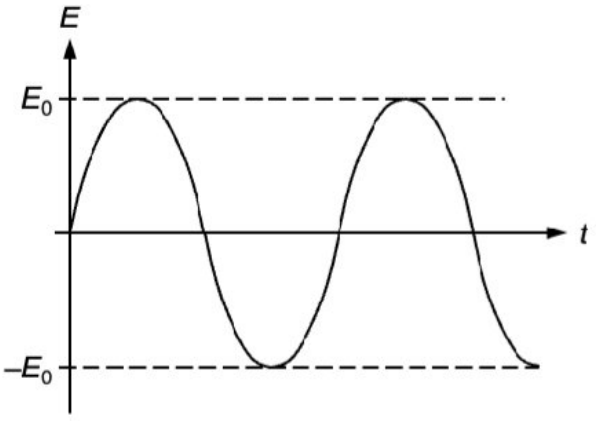
26 A coil with N turns is placed in a uniform magnetic field of flux density. The area of the coil is initially perpendicular to the field, as shown in diagram (a).



(a)

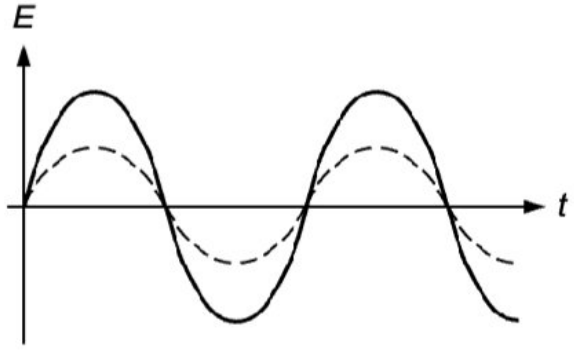


(b)

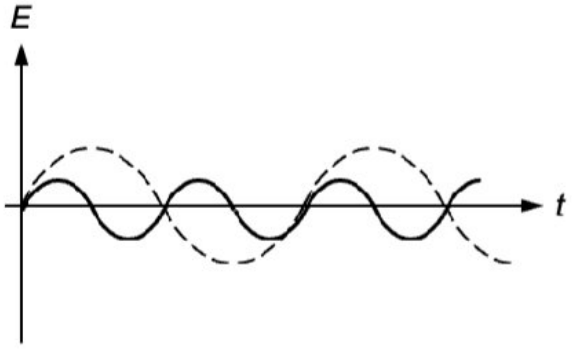
The coil is rotated around the axis and the induced e.m.f. E between the ends X and Y varies with time t is shown in diagram (b). E_0 is the magnitude of the maximum induced e.m.f.

Which diagram best represents how E varies with t if the frequency of rotation is double and the magnetic flux density remains the same? The dotted line shows the graph in diagram (b) for comparison.

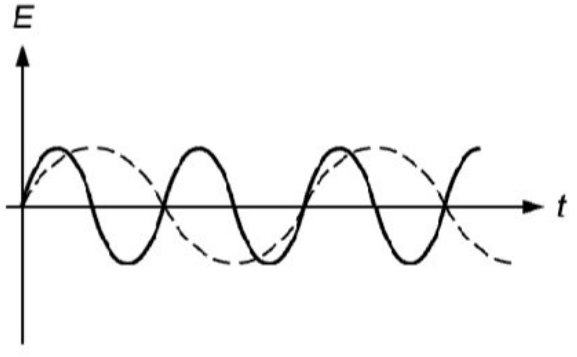
A



B



C



D

